

Title      Capacitance VS D  
Team      1  
Date      11/01/2023  
Course    NANENG 331 - Fall 2022

| D (um) | Theoretical Capacitance<br>(pF) | Simulated Capacitance<br>(pF) – Moving Substrate | Simulated Capacitance<br>(pF) – Fixed Substrate |
|--------|---------------------------------|--------------------------------------------------|-------------------------------------------------|
| 30     | 0.025794                        | 0.02313                                          | 0.02829                                         |
| 32     | 0.027312                        | 0.023074                                         | 0.029285                                        |
| 34     | 0.028829                        | 0.024984                                         | 0.030969                                        |
| 36     | 0.030346                        | 0.026105                                         | 0.032457                                        |
| 38     | 0.031863                        | 0.026921                                         | 0.033696                                        |
| 40     | 0.033381                        | 0.028289                                         | 0.034566                                        |
| 42     | 0.034898                        | 0.029534                                         | 0.036113                                        |
| 44     | 0.036415                        | 0.030963                                         | 0.03763                                         |
| 46     | 0.037932                        | 0.031984                                         | 0.038194                                        |
| 48     | 0.03945                         | 0.032825                                         | 0.040071                                        |
| 50     | 0.040967                        | 0.034418                                         | 0.041605                                        |
| 52     | 0.042484                        | 0.035032                                         | 0.043056                                        |
| 54     | 0.044001                        | 0.036259                                         | 0.044246                                        |
| 56     | 0.045519                        | 0.037608                                         | 0.045584                                        |
| 58     | 0.047036                        | 0.03792                                          | 0.046681                                        |
| 60     | 0.048553                        | 0.039395                                         | 0.048342                                        |
| 62     | 0.05007                         | 0.041105                                         | 0.04993                                         |
| 64     | 0.051588                        | 0.041164                                         | 0.050886                                        |
| 66     | 0.053105                        | 0.042715                                         | 0.052413                                        |
| 68     | 0.054622                        | 0.044112                                         | 0.053847                                        |
| 70     | 0.056139                        | 0.045432                                         | 0.055406                                        |
| 72     | 0.057657                        | 0.046393                                         | 0.056311                                        |
| 74     | 0.059174                        | 0.04763                                          | 0.057834                                        |
| 76     | 0.060691                        | 0.048766                                         | 0.059249                                        |
| 78     | 0.062208                        | 0.049446                                         | 0.060247                                        |
| 80     | 0.063726                        | 0.050518                                         | 0.061548                                        |
| 82     | 0.065243                        | 0.051918                                         | 0.063101                                        |
| 84     | 0.06676                         | 0.05275                                          | 0.064209                                        |
| 86     | 0.068278                        | 0.05417                                          | 0.065666                                        |
| 88     | 0.069795                        | 0.055979                                         | 0.067219                                        |
| 90     | 0.071312                        | 0.055745                                         | 0.067886                                        |
| 92     | 0.072829                        | 0.057343                                         | 0.069802                                        |
| 94     | 0.074347                        | 0.058472                                         | 0.071244                                        |
| 96     | 0.075864                        | 0.059804                                         | 0.072938                                        |
| 98     | 0.077381                        | 0.061299                                         | 0.07242                                         |
| 100    | 0.078898                        | 0.062303                                         | 0.07545                                         |